MOHAMED OUCHARROU

Full-Stack & AI Engineer | TypeScript, React, PostgreSQL | LMS & Multi-tenant SaaS

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SUMMARY

Final-year (5th year) Software Engineering & Artificial Intelligence engineering student and full-stack developer since 2023. Designed, shipped and maintain several **LMS platforms in production**, including **multi-tenant SaaS** with Stripe billing, RBAC and analytics dashboards. Most recently built **ArkonLMS**, a multi-tenant LMS with a **RAG-based AI layer**: multilingual pipeline on PostgreSQL/pgvector, hybrid search with RRF fusion, grounded answers with citations and a streaming student copilot.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, PHP, Java, C#
Frontend: React, Next.js, Vue.js, TanStack Router, Tailwind CSS, HTML/CSS
Backend: Node.js, Bun, ElysiaJS, NestJS, Express, REST APIs, multi-tenant SaaS architecture
AI: RAG, embeddings, vector and hybrid search (pgvector), LLM integration, SSE streaming
Databases: PostgreSQL, MySQL, MongoDB, Supabase, Drizzle ORM
Tools & integrations: Stripe, LTI, Dialogflow, Git/GitHub, Docker

PROFESSIONAL EXPERIENCE

Full-Stack & AI Engineering Intern — Arkon Systems LLC

July 2026 – September 2026 | Remote

- Designed and built **ArkonLMS**, a multi-tenant LMS (manager, teacher, student) with an AI layer based on **RAG**
- Built a **multilingual RAG pipeline** (Arabic, French, English): course material chunking, embeddings stored in **PostgreSQL/pgvector**, hybrid search with **RRF fusion**
- Added a relevance filter and **citation extraction** so answers stay grounded in the sources and hallucinations are limited
- Shipped a **student copilot** with streaming answers (SSE) and quiz generation grounded in course content, plus teacher tools for assisted course and quiz generation
- Stack: TypeScript (monorepo), Bun, ElysiaJS, React, PostgreSQL, pgvector, Drizzle, Docker

Front-End Developer Intern — NeuralMobi Ltd.

July 2025 – October 2025 | Remote

- Developed the **FanRadars** web administration interface with **Vue.js**
- Implemented user rights and roles management (**RBAC**) and the administrative features around it
- Optimized admin-side performance and security; integrated **REST APIs** for efficient data management

Freelance Full-Stack Developer

2023 – Present | Remote

- Designed and maintain several **LMS platforms in production**
- Delivered **multi-tenant SaaS** products with **Stripe billing**, RBAC and analytics dashboards
- End-to-end ownership: requirements, architecture, development and deployment

Desktop Developer Intern — ATLAS TIMNAY

May 2024 – July 2024 | Morocco

- Developed a **Point of Sale (POS)** system for a retail business; designed the database schema and the cashier interface

Web Developer Intern — SHOTCOM

July 2023 – August 2023 | Morocco

- Developed a hotel website with an **online booking module**: availability search, booking workflow and admin back-office

PROJECTS

ArkonLMS — LMS with RAG-based AI layer

TypeScript, Bun, ElysiaJS, React, PostgreSQL, pgvector, Drizzle, Docker

- Multi-tenant LMS with grounded Q&A;, **citations**, **SSE streaming** copilot and AI-assisted course and quiz generation

EduLMS — Modern LMS platform

Next.js, Supabase/PostgreSQL, Express.js

- Course management, real-time progress tracking, **multi-tier roles** and an assessment engine with automatic grading

Point of Sale system

C#, .NET, SQL

- Desktop POS with database schema design and cashier interface

EDUCATION

State Engineering Degree, Software Engineering & AI — Faculty of Sciences and Techniques, Errachidia | 2024 – 2027 (ongoing)

University Diploma of Technology (DUT), Computer Science — Higher School of Technology (EST), Meknes | 2022 – 2024

LANGUAGES

English (Professional) · **French** (Fluent) · **Arabic** (Native)